

experienced and family-centric. We are not looking particularly for those in Silicon Valley. Instead, we are on a move to make open source development very simple for everyday developers around the world, including India.

McWhirter: As the majority of developers are not the superstar variety, who work on their projects in the nights and during weekends, we designed our program accordingly. Developers generally try to get their job done. Thus, we don't want to waste their precious time while fulfilling their requirements.

Ploski: India is already one of the leading countries for those developers who want to work on their projects on a daily schedule. Thus, we are positioning our program as a key to solve challenges for Indian developers, in addition to the developers worldwide. India's traffic to our Web properties has always been No.1 or No.2. Outside of the US, it's the highest rate. Indian cities like Bengaluru, Chennai and Hyderabad are among the top-ten cities worldwide for our traffic.

Q What is hindering Indian developers to opt for open source development?

McWhirter: It's not the case of India alone; it has been so all across the world. The number of people contributing to open source is so much smaller than the people utilising open source. This gap exists even in North America. There are several large companies that use a lot of open source solutions, but it is very hard for them to spend their time in maintaining or improving open source with some new changes. I see this trend changing slowly. However, people don't have time to contribute back to open source as it's not their job. Some specialised companies allow their developers to spend some time giving back to the open source community. The open source ecosystem is okay with all this. That being said, we would love to see more people contribute back to open source developments.

It is also important how you define contribution. If I run different open source projects for my community and that community gives me feedback, it works as a contribution to me. It helps to guide me about the things I need to improve in my projects, and how.

The next step of the layer is a bug report. If no one reports bugs, then there is no value in your product and you won't be able to improve it. But if there is a bug

report, you can fix that bug and make your existing product better for its users. This not only improves the product but also acts as a great contribution.

For example, we created software that could be used in ten thousand different ways, but no one ever used it in so many ways. Using the bug reports, we got to know they used it in just 15 ways, and those were the most important ones to focus on. So bug reports and general feedback are great contributions.

We have a lot of people from India who contribute at that level. A large number of Indian open source users are using different solutions and find problems that help us improve our presence. There is probably more contribution coming from India than one can imagine. It just depends on how you define contribution.

Ploski: It is certainly a broad range. At one end you see large corporations identifying a particular project to fulfil their goals, and approaching Red Hat to be able to improve that project. It helps them make sure that the project will

survive in the long term and can be successful moving forward. That's how we receive more upstream contributions from India and global regions.

Q How is Red Hat expanding the OpenStack platform?

Ploski: We are currently the No.1 contributor to

OpenStack, which actually sees quite a bit of adoption from all kinds of industries. Specifically, telcos are planning to deploy their NFV (network functional virtualisation) on top of Red Hat's OpenStack. Our vice president of engineering, Paul Cormier, firmly believes that this is part of the future -- to be able to have your own Infrastructure-as-a-Service (IaaS) that you can have either openly, or you can have a hybrid architecture with data centres and use the public cloud.

Q Security is a big concern for the open source world. What does Red Hat offer to secure open source developments?

McWhirter: Linus' Law says, "Given enough eyeballs, all bugs are shallow." You could find a bug which may just be something that doesn't behave correctly, or a bug that breaks your entire software or steals your money. Security problems are primarily based on functional issues. There are some bugs but no one sees them until one guy figures it out and then he's got the keys to the kingdom.

“If I work in the closed source space, producing just closed source solutions, all I have to show for my work is my resume. But if I am contributing to open source, I can point to some of my repositories on GitHub and say I did this or I contributed to that.”